

Adapting Sports Nutrition Guidelines for Ultrarunning Events

Using the recommendations and strategies discussed earlier in this book, most ultrarunners can consistently fuel their workouts and maintain a good hydration status in training. It is during events that nutrition and hydration strategies frequently go off the rails. Gastrointestinal symptoms like nausea and vomiting are the top reasons athletes cite for dropping out (Hoffman 2011), and even athletes with in-depth knowledge and experience with the nutrition challenges of ultramarathons get knocked out of events by dehydration and upset stomachs.

Ultra-Specific Challenges of Ultradistance Events

One of my colleagues at CTS boiled the ultramarathoner's nutrition strategy down to this: develop the ability to simultaneously run and eat as much as you can without barfing. While that is a drastic oversimplification of the goals and challenges of fueling yourself through an ultramarathon, it is nonetheless true. It is especially true at the 100-mile distance, where caloric expenditure far exceeds the body's stored carbohydrate levels. You have to consume a lot of calories and a lot of fluid over the course of many hours in order to finish an ultramarathon, and you have to overcome several challenges that competitors in shorter events do not face.

PROLONGED DURATION

Challenge.

The simple fact of being outdoors, awake, and on your feet for many, many hours creates a nutritional challenge. Athletes in shorter events can eat a pre-race meal, consume calories and fluid to support their energy expenditure and thermoregulation needs, and then finish their event before needing another meal. They can deplete their energy stores faster and dig deeper into their reserves because postrace replenishment is only a few hours away. They can even race themselves into significant dehydration and gastrointestinal distress and reach the finish line before experiencing a significant decline in performance. Ultradistance competitors cannot be so cavalier or take as many risks because the duration and distance of the event will force them to deal with the consequences well before they reach the finish line.

Adaptation.

Ultramarathon runners must develop the ability to consume and digest a steady stream of calories over the course of many hours. Some of those calories need to be from solid foods with more bulk, more fat, and more protein than carbohydrate-only foods like gels and sports drinks. While the nutritional focus needs to be primarily on fueling with carbohydrate calories, it is important to realize your race foods are also taking the place of meals

during longer events. Fat and protein are satiating, meaning they stave off feelings of hunger better than carbohydrate does. If you respond to feelings of hunger by overconsuming carbohydrate, you increase the likelihood of gastric distress. By incorporating solid foods containing fat and protein, you can continue to base your carbohydrate intake on your caloric expenditure rather than on a growling stomach.

LONG DISTANCES/TIMES BETWEEN AID STATIONS

Challenge.

Major urban marathons have aid stations nearly every mile. That's twenty or more aid stations in twenty-six miles, with food and fluids available at the finish line. At the Western States 100-Mile Endurance Run, there are twenty-one aid stations in 100 miles, with distances between stations ranging from five to seven miles in the first third of the race and from three to four miles in the final third. Depending on your pace and the terrain, that can mean well over an hour between aid stations. If you have trouble between aid stations that causes you to stop, that time can increase dramatically.

Adaptation.

With long distances between aid stations, you will require gear that allows you to fuel along the way. Train with the gear you plan to use on race day. If you rarely run with a hydration pack in training and then have to run with one during a 100-miler, it may become uncomfortable and a constant source of annoyance. The same goes for hydration bottles or that fuel belt you just picked up online. Some races mandate certain gear that you must carry at all times. Make sure you can fit it into your pack, that the pack is still comfortable, and that you can run effectively when the pack is fully loaded. If you will have pacers or support crews in aid stations, make sure they know how to quickly and efficiently refill your pack so you don't have to stop and wait.

DRAMATIC CHANGES IN ENVIRONMENTAL CONDITIONS

Challenge.

If your event starts in the dark and proceeds through an entire day and into the night, and perhaps even into the following morning, you are going to experience dramatic shifts in environmental conditions. Ultramarathon runners can face temperature swings of forty or more degrees and high-altitude rain or snowstorms (even in summer). These changes greatly impact your sweat rate and your perceptions of hunger and thirst.

Adaptation.

Use your equipment to minimize changes in your core temperature due to changes in environmental conditions. If it's going to get hotter, you want to take steps to stay cool: wear lighter clothing, remove layers, soak your clothing, drink cool beverages, and so forth. If it is going to get colder, use your equipment to stay warm by wearing more layers and swapping wet clothing for dry clothing. In both situations, you need to stay hydrated. As weather changes, be aware of changes in your sweat rate because that directly impacts your hydration strategy. As temperatures climb, you need to anticipate an increase in your sweat rate and start consuming more fluid. The longer you wait, the more behind you get in hydration status. In cold weather, your sweat rate will decline; it's important to recognize that and reduce fluid intake to avoid overhydration. If you are behind in hydration already, you can also take advantage of cooler temperatures to gradually get your hydration status back on track. If you anticipate dramatic changes in temperature during your event, know your sweat rate for a variety of temperature ranges. Dramatic environmental changes also disrupt your routine and strategy. You may forget to eat as you struggle through a thunderstorm or focus on staying warm through the predawn hours. This is where practice and a knowledgeable support crew can be invaluable.

GENERAL FATIGUE AND DIMINISHED DECISION-MAKING ABILITY

Challenge.

When you are very tired, you make decisions you would never make in a less-fatigued state. You might leave an aid station without refilling a hydration pack. You might zone out and not consume anything for miles. You might scrap your nutrition plan entirely and scarf down enough cookies to make you sick. I've seen all these things and many more, all caused by extreme fatigue.

Adaptation.

Habits and routines pay huge dividends for an ultradistance competitor. Develop habits that are consistent from everyday training sessions to your longest races so they become ingrained to the point that you stick to them even when you are unbelievably fatigued. Develop a routine you use every time you pick up your hydration pack: check for water, standard food choices, and standard equipment (like a rain shell). Make sure you at least have your bull's-eye foods (more on those later) with you whenever you leave an aid station. Keep these routines simple and minimal; you don't have to have everything all the time, but you always need to have something to eat, something to drink, and something to protect yourself from the elements. When you are fatigued and in the heat of battle, have a routine.

FOOD FATIGUE

Consuming small amounts of food and fluids over and over again during the course of many hours is exhausting. Many athletes reach a point where they don't want to expend the effort required to eat or they no longer have a taste for the food options available. This is rarely a problem in shorter events because less food is needed to complete the race. As a result, athletes competing in shorter events can utilize a narrower range of food options, whereas ultradistance athletes typically need to find a wider range of foods they are willing to eat and that work for them without causing an upset stomach. I have spent so much time working through the challenge of food fatigue with athletes that I have developed a specific strategy to deal with it.

The Bull's-Eye Nutrition Strategy

During training, particularly during longer runs, I have athletes experiment with different—and sometimes counterintuitive—foods to help them reach their target calorie ranges. Foods that work are easy to open and eat on the run. They taste good, don't get stuck in your teeth, and make you run as well as or better than you were running before eating them. Foods that don't work are difficult to open, messy, crumbly, and hard to hold in one hand. They get stuck in your mouth, are too dry, or are tough to swallow without choking. Most of all, they sit in your gut like a calorie bomb, make you feel bloated or full, and slow you down. Your goal is to find three to five foods you can count on to work in any situation. These are your bull's-eye foods.

All the food options you try can be categorized by where they fall on a target (picture an archery range—that kind of target). Your bull's-eye foods are your tried-and-true favorites. If these core foods begin to fail because you're tired of eating them, craving more sweetness or saltiness, or craving a different texture, then you can choose foods from the next ring of the target. These are foods you may not eat all the time, but you have tried them in training and know they work for you.

Beyond this ring are foods that you haven't tried but that are similar to foods you have tried. For instance, you may know that chocolate chip cookies work for you, but there are only oatmeal cookies in the aid station. Or you like regular potato chips, but only BBQ-flavored chips are available.

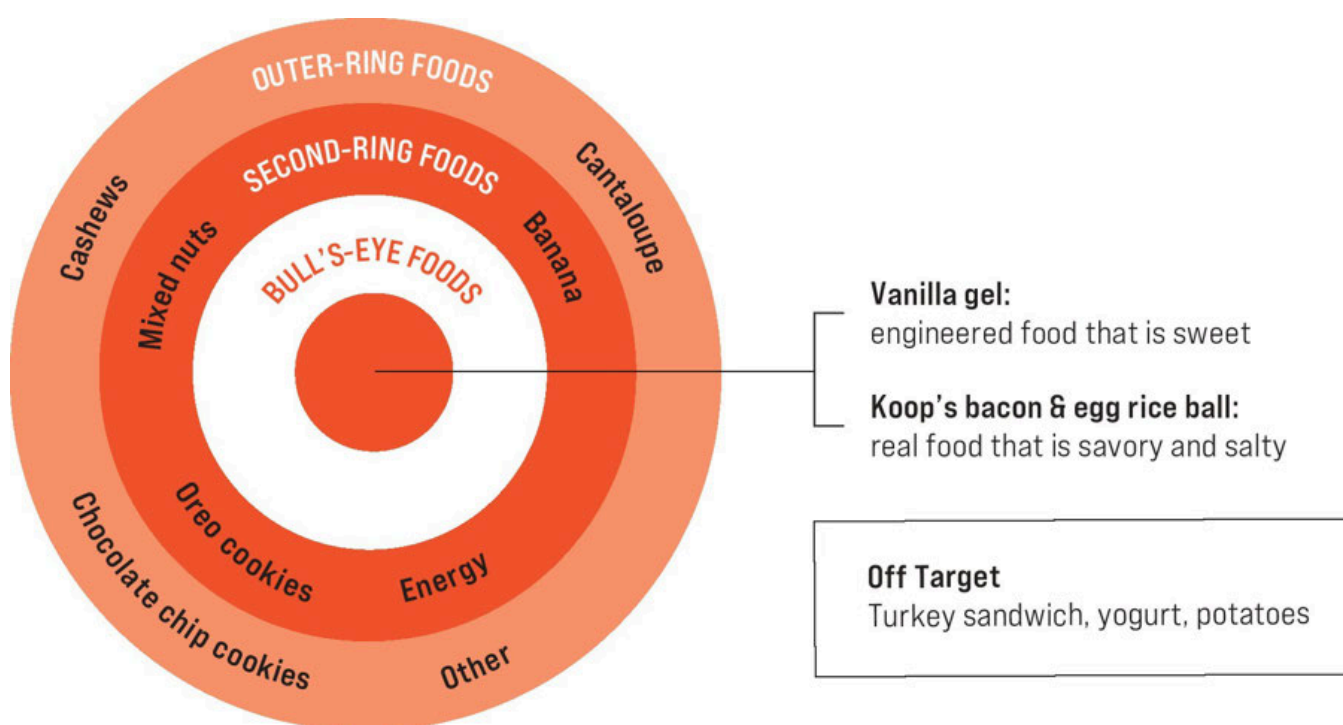


FIGURE 14.1 A sample bull's-eye nutrition strategy.

Anything beyond this ring is off target altogether. These are the foods you know don't work for you and foods similar to foods that don't work for you. It is important to list these foods out, as well, so that you and your crew are reminded of the things you have tried that have not worked in training.

Developing Your Bull's-Eye Foods

Variety is important in your short list of bull's-eye foods. The end goal is to find a combination of three to five foods that, together, meet all the following criteria:

- **AT LEAST ONE REAL FOOD** —something you make or assemble or that is not made specifically for running (rice ball, peanut butter and jelly sandwich, or pretzels)
- **AT LEAST ONE ENGINEERED FOOD** (gel, chewable, or sports drink)
- **SOMETHING SWEET**
- **SOMETHING SAVORY**
- **SOMETHING SALTY**

If you construct this combination correctly, these bull's-eye foods can be rotated and substituted during any race as needed, according to your target calorie range, which you identified earlier in this book. After these core foods have been fully vetted, experiment with backup (second-ring) foods. These backup foodstuffs are what you can confidently fall back on if you lose your taste or craving for your bull's-eye foods. The typical fall back plan revolves around the aid station fare of cookies, soup, fruit, and sandwiches. Outer-ring foods will fall into this aid station assortment as well.

The bull's-eye strategy is easy to visualize and easy to explain to your support crew. Figure 14.1 is an actual example of one runner's bull's-eye strategy. At any point during a race, either the athlete or her support crew can quickly consult the target and make a good decision.

What about Strategies to Enhance Fat Metabolism?

Fat adaptation, or the ability to utilize more fat as fuel at higher intensities, has been a hot topic in the endurance sports world for the past decade. Some of the best minds in exercise science have dedicated the better parts of their careers to researching and writing seminal papers on the topic and helping athletes perform at their very best (see Louise Burke, Asker Jeukendrup, and others). Considering that gastrointestinal distress is one of the primary villains stopping ultrarunners in their tracks, the idea of becoming a better fat-burning machine is attractive. Additionally, the demands of ultramarathon racing involve running at lower intensities for long periods of time, which utilizes fat for energy to a greater extent than races like the 10K and marathon. Since you have a virtually unlimited source of energy stored in your body as fat and all the opportunity to use it at ultramarathon intensities, it makes sense to want to exploit this metabolic confluence of events. We also know that substrate metabolism is trainable and that with changes to diet and training, rates of fat oxidation can be as much as doubled for a given submaximal intensity (Purdum et al. 2018). However, what sounds like a relatively attractive strategy on the surface is a little less straightforward in practice.

A misnomer in the “fat-adapted” athlete world is that the only way to become more efficient at utilizing fat is by following a ketogenic, or extremely low-carbohydrate high-fat (LCHF) diet. The reality is that your body becomes fat adapted through a few different conduits. Normal endurance training improves fat oxidation by increasing your body's ability to produce energy aerobically. As you become more fit, you can utilize more oxygen at any given

intensity, and thus you have a greater opportunity to utilize fat as a fuel. Dietary interventions such as a LCHF or ketogenic diet can further enhance this opportunity by upregulating fat oxidation pathways. You can also alter substrate metabolism to prioritize fat by changing fuel availability during specific training sessions. All of the aforementioned methods are effective. Remember, your body is smart. It is already a flex-fuel machine capable of burning multiple fuel sources and upregulating the fuel utilized based on training status and availability.

For many, optimizing fat metabolism to improve performance seems to be a win-win. You win by being able to tap into a nearly limitless fuel source and at the same time not rely on as much exogenous carbohydrate. You'll be more self-sufficient, less reliant on gels and aid station foods, and ultimately stave off gastrointestinal distress.

With so much going for it, what downsides could there possibly be from following an optimized approach to better fat utilization? The simplest answer is that you are compromising your ability to perform at higher intensities. This is because it takes approximately 8 percent more oxygen to liberate energy from fat as it does from carbohydrate. While fat is a nearly limitless energy source, it is also a more expensive fuel to use. Some make the argument that the increased oxygen cost of liberating energy from fat is offset or justified by the fact oxygen delivery is at less of a premium at lower intensities, so the oxygen penalty for relying on fat for fuel is seemingly less important when running at 50 to 60 percent of your VO₂ max, as you have some bandwidth to spare. But it's not just less economical; it limits your capacity to utilize other substrates.

When you manipulate your diet in an attempt to optimize fat metabolism, carbohydrate becomes scarcer as a fuel source and your body also becomes less efficient at utilizing carbohydrates (Burke et al. 2016; Jeukendrup 2017). While theoretically a good enough runner could run and walk slowly enough to complete 100 or even 200 miles on energy from stored fat alone, for mere mortals, even in the most fat-adapted state, ultramarathons are long and intense enough that athletes will need some exogenous carbohydrate. So yes, after a few weeks on a lower-carbohydrate diet you might now be a fat-metabolizing machine, but without the introduction of carbohydrates during and around training leading up to your event, you might not be able to effectively transport the carbohydrates you are trying to ingest (even in lower amounts), ironically making you more likely to experience gastrointestinal distress.

The economy penalty of relying on fat for energy doesn't just hurt race performance; training suffers, too. Reducing economy reduces performance, particularly at higher intensities. This means the amount of high-intensity work you can do is limited, either by a reduction in intensity (running slower during your intervals) or a reduction in the time you can spend at any intensity, or both. In any form, this reduces the training stress you can get out of an interval session like TempoRuns or RunningIntervals and therefore the benefits you can reap from those sessions. Furthermore, some training adaptations are attenuated in the absence of carbohydrate, meaning you quite literally get less bang for your hard-earned training buck.

Finally, a more recently explored drawback of following a LCHF strategy is the potentially increased risk of bone-stress injuries. In a first-of-its-kind study looking at LCHF diets and bone health, researchers found that even after three and a half weeks of a LCHF diet athletes showed changes in critical markers of bone health (Heikura et al. 2020). Specifically, subjects showed increased levels of CTX, a marker of bone resorption, and a decrease in P1NP, a marker of bone formation. These negative adaptations may have implications for injury risk and long-term bone health.

As with other areas of training and sports nutrition, fat adaptation is neither all good nor all bad. From a performance standpoint, I do not see compelling evidence for recommending a full-time LCHF nutrition strategy for ultramarathon runners, but I also understand that athletes make dietary decisions for a lot of reasons. If you choose to be a LCHF ultrarunner, that's fine as long as you accept the performance limitations that result from that decision. Alternatively, you can manipulate your diet to achieve specific goals.

DIETARY MANIPULATION

Dietary manipulations to enhance fat metabolism involve controlling the amount of carbohydrate you consume on a given day. These manipulations come in three main flavors:

DIETARY STRATEGY

CARBOHYDRATE CONTENT

Very low-carbohydrate ketogenic diet

<50g carbohydrate/day

Low-carbohydrate diet

15–30% of calories from carbohydrate

High-carbohydrate diet

60–65% of calories from carbohydrate

Consensus from International Society of Sports Nutrition

60% of calories from carbohydrate

TABLE 14.1 Carbohydrate content of different dietary strategies. Adapted from Burke 2020; Tiller et al. 2019; Wylie-Rosette 2016.

- **DIETARY KETOSIS:** The most restrictive is a ketogenic diet, which leverages the body's ability to produce and utilize ketones for fuel in the absence of carbohydrate. The diet garnered attention initially as a treatment for children with severe cases of epilepsy, before it was adapted into the nutrition and sports worlds as a way of manipulating substrate utilization and controlling weight. The diet consists of 75 to 85 percent calories from fat and 15 to 20 percent calories from protein, but the crucial parameter is restricting carbohydrate intake to fewer than 50 g per day (the equivalent of two to three bananas or two energy gels).
- **LOW-CARBOHYDRATE HIGH-FAT (LCHF):** A low-carbohydrate high-fat diet is next on the plate. While definitions vary, this diet is less restrictive than dietary ketosis and consists of 30 to 40 percent calories from carbohydrate. These two strategies run contrary to most sports science recommendations that call for 60 to 65 percent of an endurance athlete's calories to come from carbohydrate (Tiller et al. 2019).
- **PERIODIZED CARBOHYDRATE:** The third option is a periodized carbohydrate (PCHO) approach. The periodized approach can represent a blend of a ketogenic diet, LCHF diet, and high-carbohydrate low-fat diet, as well as approaching specific training sessions with low carbohydrate availability (more on that next). On a PCHO diet you would leverage high carbohydrate availability by topping up muscle glycogen stores and fueling with carbohydrate during workouts for higher-intensity periods of training and/or specific workouts. Other less-intense workouts would be completed with intentionally low carbohydrate availability, either through diet manipulation, overnight fasting, or a low-carbohydrate day to delay postexercise glycogen replenishment. This "best of both worlds" strategy potentially allows you to reap the benefits of improving fat oxidation without limiting your ability to do higher-intensity work, thus maintaining quality throughout your training cycle so you are not shortchanging your fitness potential.

DIETARY STRATEGY	ADVANTAGES	DISADVANTAGES
Low-carbohydrate high-fat or Ketogenic (defined by less than 50g CHO/day)	Increased fat oxidation, sparing endogenous glucose Lower need for exogenous carbohydrates during activity Decreased body fat percentage	Inability to train and race at higher intensities Less efficient at transporting carbohydrates across gut membrane Increase risk of bone-stress injuries from hormonal changes affecting bone remodeling Restricted sources of food, specifically fruits and vegetables Can lead to low energy availability
High-carbohydrate low-fat	Consistently high training quality Trains the gut to facilitate more glucose absorption	Possible overreliance on carbohydrate as a fuel source
Periodized carbohydrates	Matches training intensity/duration to substrate needs Enhanced fat oxidation due to cellular changes in the muscle No diminished training quality	Logistically difficult to implement

TABLE 14.2 Advantages and disadvantages of dietary strategies to manipulate substrate utilization.
Adapted from Burke 2020; Tiller et al. 2019; Wylie-Rosette 2016.

TRAINING MANIPULATION

Training manipulations for improving fat oxidation focus primarily on restricting carbohydrate availability for specific training sessions. The two primary strategies employed are:

1. Training twice per day and withholding carbohydrate intake after the first session. In this case, the first session of the day is a harder interval workout to deplete glycogen stores and the second session of the day is an EnduranceRun or RecoveryRun.

2. Fasted runs. In this case, the run starts with lower carbohydrate availability either due to an overnight fast (if the run is performed in the morning) or due to moderating carbohydrate throughout the day (if the run is performed in the evening). The training session is typically limited to less than two hours and performed without fuel.

Both of these strategies work on the premise that the low-carbohydrate training session amplifies your body’s cellular signaling to produce more fat oxidizing machinery. This includes making more mitochondria as well as upregulating oxidative enzymes (Marquet et al. 2016).

Now all this feels a little bit like goldilocks and the three four bears. We want to find a situation that (1) leverages our fat-burning capabilities so we can utilize this nearly limitless yet expensive source of energy, (2) maintains performance at higher intensities, (3) promotes adaptation from all of our hard work in training, and (4) avoids any deleterious effects to bone health stemming from low energy availability. Is there a happy medium? It depends on who you ask. But since you are reading my book, I’ll give you my humble opinion.

TRAINING STRATEGY	STEP 1	STEP 2	STEP 3	STEP 4
Two-a-day	Running interval session	Restrict carbohydrate immediately post run	Second session of 1.5–2-hour EnduranceRun performed without fuel	Refuel with carbohydrate post run
Fasted run	Overnight fast	EnduranceRun of 1.5–2-hour upon waking. Performed without fuel	Refuel with carbohydrate post run	

TABLE 14.3 Training strategies to enhance fat oxidation.

SHORT-TERM ENERGY DEFICITS AND RED-S

With any of these strategies, you are going to have to go through at least some shortterm energy deficits, where you’re consuming fewer calories than required for effective training. You might think, “No biggie, I’ll just grab some more pizza for dinner to make up the balance.” However, we’re learning more and more that RED-S is problematic for endurance athletes. This normally happens when there is an energy defi cit caused by a workout combined with and compounded by underfueling. Fat-adaptation strategies can further compound the negative effects, as they rely on short- or long-term energy deficits. The host of performance complications include decreased muscle strength, decreased performance, increased injury risk, and decreased training response, to name a few. In this sense, the negatives of most of the fat-adaptation strategies can quite quickly outweigh the positives.

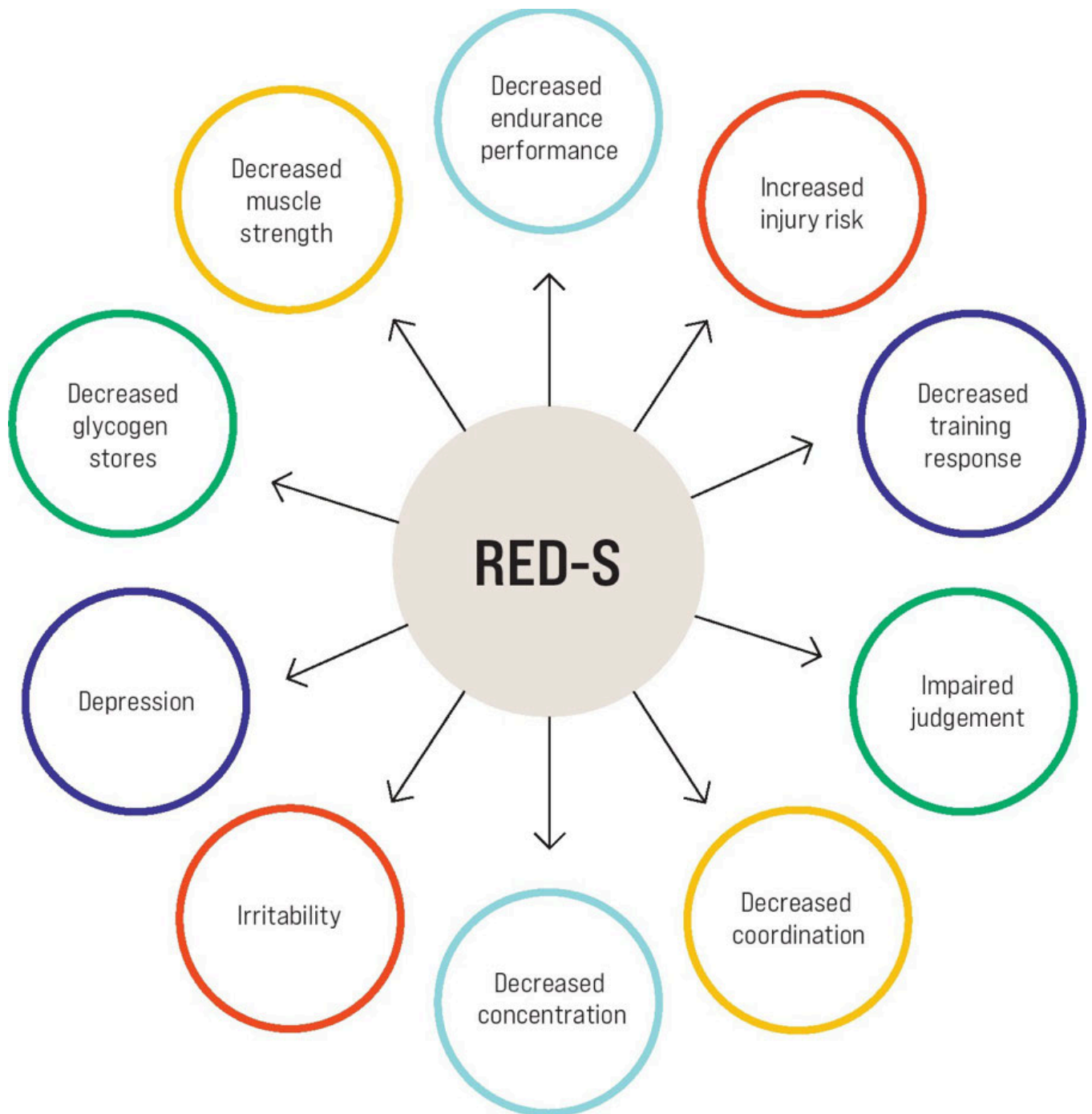


FIGURE 14.2 Effects of RED-S. Adapted from Mountjoy et al. 2018.

ULTRAMARATHONERS ARE ALREADY REALLY GOOD AT BURNING FAT

Simply by running copious hours at a relatively low intensity, ultrarunners are already really good at burning fat. As stated in the 2019 IAAF consensus statement on nutrition for athletes: “high fat oxidation rates also appear to be inherent in ultramarathon runners regardless of background macronutrient dietary modifications.” Trying to double down on an effect you already have is a high-risk, low-reward (or no-reward) scenario.

ADHERENCE IS PROBLEMATIC, METABOLIC MANIPULATION IS TRANSIENT, AND EXPERTISE IS SCARCE

With any of these fat-adaptation strategies, adherence is difficult. We know from the work on the ketogenic diet that you need at least a few weeks of adherence before maximum benefits of fat oxidation start to appear. And after all that, the enhanced fat-burning effect loses its potency within a few days of reintroducing carbohydrates. While less studied, the transient nature of other fat-adaptation strategies is likely similar: they will take some time to show an effect, and that effect disappears quickly. Additionally, safe implementation requires technical knowledge from a registered dietitian who can manage overall energy and micronutrient balance alongside the intended metabolic manipulation. Your fat-adapted friend and the athletes promoting these diets do not qualify as experts in this area, no matter their level of enthusiasm. Quite frankly, the level of expertise required is scarce and many athletes will not have access to the right support system.

WHAT ABOUT ATHLETES WHO ARE CRUSHING IT ON LOW-CARB DIETS?

It would be remiss of me not to discuss the numerous well-known and not-so-well-known athletes who have been able to improve their performances with fat-adaptation strategies. They are out there for sure, filling up your Instagram and Strava feeds with their latest accomplishments.

I will let you in on a not-so-secret: almost all of these performance benefits have nothing to do with the fat-adaptation side of the equation. They have almost entirely to do with the fact that the athlete lost weight. This point is driven home when I hear athletes advocate for using a ketogenic diet as a "tool". What this really means is that it's a tool for energy restriction, and the resulting weight loss leads to improvement.

I DON'T THINK IT'S A GOOD HEALTH MESSAGE

Finally, if you are spending lots of time figuring out how or where to skip your gel (or meal for that matter), you've got it all wrong. Much of the issue I have with fat-adaptation strategies lies in the fact that there is too much fixation on what you are leaving out (breakfast, for example) and not enough on what you are putting in. If you want to be an athlete, the focus needs to be on the latter. High-quality foods, not the absence of them, are what athletes should focus on.

We need less obsession over calorie composition and more over calorie quality and adequate quantity.

When you focus on quality and adequate quantity over composition and restriction, the results are physically, emotionally, and psychologically healthier outcomes (see Figure 14.2 for psychological/emotional complications stemming from RED-S).

There is no greater example of poor judgement caused by intentionally fat adapting than the term "strategic tuber" or "strategic fruit," which I have seen pop up in the ketogenic and low-carb community. "Strategic" in this sense fits the narrow definition of a way of consuming carbohydrates (tubers and fruit in this case) to fuel intense workouts, but not so much that it violates some predetermined principle of carbohydrate restriction. Since when do we need to "strategically" use a food source that comes from the ground and is rich in vitamins and nutrients, tasty, and fulfilling? If a simple "not-so-strategic" banana—with all its vitamins and minerals—is going to destroy your nutrition strategy, there's something wrong with your nutrition strategy, not with the banana.

Just eat the sweet potatoes, bananas, and apples, people. They're good for you.

THERE IS ROOM FOR PERIODIZING CARBOHYDRATE AVAILABILITY AS WELL AS SESSIONS WITH LOWER CARBOHYDRATE AVAILABILITY, WITH A TON OF CAVEATS

If you want to double down on your body's natural fat-burning machinery, there are a couple of low-risk, medium-reward tools in your tool belt. First, you can periodize your macronutrient composition so that during your EnduranceRun and SteadyStateRun phases, you are consuming ~50 percent of your daily calories from carbohydrate. For any TempoRun or RunningInterval phases, you should up this intake to ~60 percent. This is akin to the PCCHO approach mentioned earlier. Second, doing an early morning EnduranceRun of up to two hours after an overnight fast is likely another low-risk strategy. Finally, a two-a-day strategy where you perform an interval session (like TempoRun or RunningIntervals) with adequate carbohydrate availability followed by an EnduranceRun with lower carbohydrate availability can help promote fat oxidation without many deleterious effects, provided that you can carve out the time in your day for two sessions and the extra logistical work does not pile on additional nonworkout stress.

AVOID THESE THINGS

I realize that many of you reading this book will still be tempted by aggressive fat optimization strategies. I get it. As I mentioned in the onset of this section, on the surface it seems an attractive proposition. If you do decide to undertake one of these, please avoid the following at all costs:

1. Low energy availability. Taking in too few calories, regardless of their macronutrient composition, will lead to a whole host of problems. Remember, with any fat-adaptation strategy the goal is to manipulate one macronutrient's availability, not total caloric availability.
2. Hard workouts in a fasted state or while restricting carbohydrate. Seriously, this is a terrible idea. You set yourself up for underperformance and overtraining all at the same time.
3. Hard workouts at the end of long runs. For whatever reason, I have seen this creep up from time to time in athletes' training, either as a means of "training on tired legs" or some sort of fat-adaptation strategy. Just don't do it. Like point #2 above, you are taking on a boatload of stress for little positive outcome.
4. Long runs without fuel. Don't go out and do your longest of long runs without fuel. A very long fasted run does not produce superior adaptations compared to some of the more reasonable strategies mentioned above. So, these activities have very little, if any, upside and a whole lot of downside. Sure, you can "get through it" by being tough, running slowly, and hydrating properly, but the reward for your effort is nil and might even result in maladaptation. The laurels of ultrarunning are littered with tales of well-intended athletes running 50 or even 100 miles without fuel and then wearing that accomplishment as a badge of honor. Last I checked, there are no awards for the lowest number of calories consumed in a race.

Balancing Hydration and Sodium: The Ultimate Challenge for Ultrarunning

Your total body water and your sodium concentration will affect performance in different ways. Further complicating the matter is the fact that at any point in time, your hydration and sodium balance can be normal or you could have too much or too little of either one.

Hydration state can be measured relative to your body weight, and your natremic state is measured by the concentration of sodium in your blood (measured in millimoles per liter). To understand how hydration status and sodium concentration change, it's important to first understand some terminology.

- EUHYDRATED: Normal level of hydration, typically less than 3 percent weight change
- HYPOHYDRATED/DEHYDRATED: Greater than 3 percent decrease in body weight
- HYPERHYDRATED/OVERHYDRATED: Increase in body weight
- NORMONATREMIC: Normal blood sodium concentration (135 to <145 mmol/L)
- HYPONATREMIC: Too little sodium in the blood: a blood sodium concentration of <135 mmol/L
- HYPERNATREMIC: Too much sodium in the blood: a blood sodium concentration of >145 mmol/L

[Table 14.4 helps us to better visualize how these hydration and natremic states converge. It resembles a tic-tac-toe board, with the various hydration states on the horizontal rows and natremic states in the vertical columns. As with regular tic-tac-toe, occupying the center square puts you in a position of power. This is the desired physiological state in which you can run faster, run farther, and tolerate heat better.](#)

Assuming you start your event euhydrated and normonatremic, during the course of an ultramarathon you are going to consume a lot of fluid, a lot of calories, and a lot of sodium. You will also be exposed to varying temperatures and weather conditions, and you will spend times at a variety of intensities. All these things, along with the sheer length of time you will be out on the course, will shift your hydration and natremic statuses.

No scenario is more advantageous than being euhydrated and normonatremic, but some of the other scenarios are worse than others. The one that gets the most media attention is the scenario in which an athlete is overhydrated and hyponatremic, specifically a slow marathoner who drinks way too much plain water and gains weight during the event. Hyponatremia happens in ultrarunning, too, but it can also manifest as euhydration and hyponatremia. This means the athlete is at normal body weight (less than 3 percent body-weight loss) but has not replenished enough sodium to offset losses from prolonged hours of sweating. It's dilution of the blood by removal of sodium, rather than dilution of the blood by the addition of too much water.

		Natremic States		
		Low	Normal	High
		Hyponatremic	Normonatremic	Hypernatremic
Hydration State	Low	Dehydrated and hyponatremic	Dehydrated and normonatremic	Dehydrated and hypernatremic
	Normal	Euhydrated and hyponatremic	Euhydrated and normonatremic	Euhydrated and hypernatremic
	High	Overhydrated and hyponatremic	Overhydrated and normonatremic	Overhydrated and hypernatremic

TABLE 14.4 How hydration and natremic states converge.

During an ultramarathon your sodium-hydration balance is one of the most important aspects of biofeedback to pay attention to. As you eat, drink, run, and spend more time in the elements, this balance is constantly shifting. Your goal is to stay in or get back to that center square!

The following sections represent the various combinations of hydration and natremic states you could end up in during the course of an ultramarathon. Some are more dangerous and affect your performance more than others. Similarly, some are easier to get into and easier to correct than others. To better explain them, each state is described by the level of hydration, level of sodium, the prevalence of the occurrence (extremely rare = less than 1 percent, rare = 3 to 6 percent, somewhat common = 10 to 15 percent, common = 30 percent or greater; adapted from Hoffman, Hew-Butler, and Stuempfle 2013), how you got there, what the symptoms are, and how you can get back to the center square (euhydrated and normonatremic).

COMBINATION 1: DEHYDRATED AND HYPONATREMIC

- **HYDRATION:** Low
- **SODIUM:** Low
- **OCCURRENCE:** Rare (3 to 6 percent of 100-mile finishers)

HOW YOU GOT HERE. You can get to this state if you have lost a lot of electrolytes through sweating and you have gradually fallen behind in both sodium and fluid replenishment. Rising temperatures during an ultramarathon can lead to this problem as well, because you have been losing sodium throughout the race, and then sweat rate increases dramatically and accelerates both fluid and sodium loss.

SYMPTOMS. Dry skin and mouth, thirst, craving for salty foods, dizziness on standing, weight loss, and low/no urine volume. At risk for heat stress in warm conditions.

HOW TO GET BACK TO THE CENTER SQUARE. Consume sports drink, salty broth, and/or a combination of water and salty foods. Don't consume only water. In warm/hot conditions slow down, wet your clothing, and douse with water to alleviate heat stress.

COMBINATION 2: EUHYDRATED AND HYPONATREMIC

- **HYDRATION:** Normal
- **SODIUM:** Low
- **OCCURRENCE:** Rare (3 to 6 percent of 100-mile finishers)

HOW YOU GOT HERE. This is a relatively easy state to reach in an ultramarathon, especially if you experience food fatigue and gradually start taking in less food or fewer sodium-containing drinks. Athletes who are salty sweaters can also end up in this state because they have higher sweat losses over time. This condition does not put you at elevated risk of heat stress, since you have plenty of body fluid for sweat; however, it can become much more serious if sodium balance is not corrected.

SYMPTOMS. Normal weight and thirst, moist mouth, craving for salty foods but nausea upon eating, and normal urine output.

HOW TO GET BACK TO THE CENTER SQUARE. Consume sports drink, salty broth, and/or a combination of water and salty foods. Try ginger or another remedy to alleviate nausea.

COMBINATION 3: OVERHYDRATED AND HYPONATREMIC

- **HYDRATION:** High
- **SODIUM:** Low
- **OCCURRENCE:** Rare (3 to 6 percent of 100-mile finishers)

HOW YOU GOT HERE. This dangerous condition requires immediate action. Athletes who consume too much plain water and not enough sodium are at risk. Even those who know better can end up in this situation when food fatigue and overall fatigue cause poor decision-making. Some stop eating due to nausea

but continue consuming a lot of plain water in hopes of calming the nausea by getting the gut to start moving again. Hot conditions combined with frequent aid stations can make this worse, as sips of low-sodium fluids (cola, ginger ale, water) at aid stations add up.

SYMPTOMS. Increased weight, puffy hands, high output of clear urine, cognitive impairment, disorientation, sloshing stomach, nausea, and possible vomiting. Low risk of heat stress because there is plenty of fluid available for sweat.

HOW TO GET BACK TO THE CENTER SQUARE. If symptoms are mild and the condition is caught early, you may be able to return to a more euhydrated and normonatremic state by consuming salty foods and restricting fluid intake. Even if symptoms are mild, it is recommended that you stay in an aid station where medical professionals are present. The support crew needs to monitor the runner's condition closely because it can deteriorate quickly. A runner who is exhibiting frequent vomiting, seizures, or significant neurological impairment needs immediate attention from medical staff. When in doubt, seek medical help.

COMBINATION 4: DEHYDRATED AND NORMONATREMIC

- **HYDRATION:** Low
- **SODIUM:** Normal
- **OCCURRENCE:** Somewhat common (10 to 15 percent of 100-mile finishers)

HOW YOU GOT HERE. In this case, you are consuming plenty of sodium from a variety of foods but either are not consuming enough fluid or are underestimating your sweat rate. At high altitudes the air is dry and sweat evaporates quickly, so skin may seem pretty dry even though you are sweating profusely. You also lose more body fluid through respiration in dry air. This can lead runners to underestimate sweat rate and fail to replenish fluids adequately. In warm environments some athletes run low on fluids based on the amount they are carrying. Rationing fluid is better than running completely dry, but eventually it will lead to dehydration.

SYMPTOMS. Weight loss, dry mouth, strong thirst, normal appetite for food, possible dizziness upon standing, fatigue, loss of focus, low urine output, and urine may be the color of apple juice or darker. Elevated risk of heat stress in warm environments.

HOW TO GET BACK TO THE CENTER SQUARE. Consume water. Sports drink and/or salty broth can be consumed but are not necessary. In warm/hot conditions slow down, wet your clothing, and douse with water to alleviate heat stress.

COMBINATION 5: OVERHYDRATED AND NORMONATREMIC

- **HYDRATION:** High
- **SODIUM:** Normal
- **OCCURRENCE:** Common (>30 percent of 100-mile finishers)

HOW YOU GOT HERE. This scenario sometimes happens in cooler weather when sweat rates are relatively low and you are eating plenty of sodium-rich foods but also consuming more fluid than necessary. Sometimes athletes who follow a regimented fluid schedule will end up here because they don't reduce fluid intake when cool weather, rainstorms, and wet clothing reduce their sweat rate. However, even when the temperature rises, athletes can be susceptible to this issue. Simple paranoia about dehydration can lead them to overdrink. Research performed by Martin Hoffman during the Western States 100 and the Rio Del Lago 100 Mile Endurance Run has found that this scenario is the most common (aside from euhydration and normonatremia) for athletes (Hoffman, Hew-Butler, and Stuenkel 2013).

SYMPTOMS. Nausea, sloshing stomach, bloating, low thirst, slight weight gain or absence of expected minor weight loss, puffy hands, and normal taste for salty foods but diminished hunger due to feeling full.

HOW TO GET BACK TO THE CENTER SQUARE. Restrict fluid intake to just moisten the mouth. Consider just swishing fluid in the mouth and spitting it out. This is not a scenario that increases the risk of heat stress because there is plenty of fluid available for sweat.

COMBINATION 6: DEHYDRATED AND HYPERNATREMIC

- **HYDRATION:** Low
- **SODIUM:** High
- **OCCURRENCE:** Extremely rare (<1 percent of 100-mile finishers)

HOW YOU GOT HERE. The difference between this scenario and being dehydrated and normonatremic may be the absence of urination. With high sodium concentration in the blood, your body holds on to the water it has instead of making fluid available for urine. Athletes sometimes end up in this condition due to overconsumption of sodium through salt tablets or losing track of sodium intake from multiple sources (chips plus salt tablet plus electrolyte drink, etc.). The challenge with this scenario is that the high sodium consumption may lead to nausea, which reduces the appeal of drinking.

SYMPTOMS. Weight loss, dry mouth, dry skin, cessation of urination, strong thirst, and salty foods may taste bad. Athlete is at elevated risk of heat stress in warm environments.

HOW TO GET BACK TO THE CENTER SQUARE. Consume plain water and restrict sodium intake. If you need calories, seek low-sodium carbohydrate sources. Sip small amounts of water to mitigate nausea. In warm/hot conditions slow down, wet your clothing, and douse with water to alleviate heat stress.

COMBINATION 7: EUHYDRATED AND HYPERNATREMIC

- **HYDRATION:** Normal
- **SODIUM:** High
- **OCCURRENCE:** Extremely rare (<1 percent of 100-mile finishers)

HOW YOU GOT HERE. This state is caused by consuming too much sodium in too short a period of time. Athletes who use salt tablets when they don't need them can end up here easily, which is one of the reasons I recommend food and fluid sources of sodium rather than salt tablets.

SYMPTOMS. Normal weight, strong thirst, no dry mouth, salty foods may taste bad but appetite is generally normal, and normal urination frequency but urine is the color of apple juice or darker. Risk of heat stress is not elevated.

HOW TO GET BACK TO THE CENTER SQUARE. Restrict sodium intake and continue with normal consumption of water. If you need calories, seek low-sodium carbohydrate sources. The condition will improve as you lose sodium through sweat and urination, but be careful not to restrict sodium long enough to swing all the way to euhydrated and hyponatremic.

COMBINATION 8: OVERHYDRATED AND HYPERNATREMIC

- **HYDRATION:** High
- **SODIUM:** High
- **OCCURRENCE:** Extremely rare (<1 percent of 100-mile finishers)

HOW YOU GOT HERE. This is a scenario that's relatively hard to achieve. To get here you have to overconsume both fluid and sodium. It can happen, however, if an athlete is moving slowly in cool weather (low sweat rate) and consuming large amounts of salty food and/or large volumes of salty broth or high-sodium sports drinks. When you are moving slowly, it is easier to consume large volumes of food and fluids because the intensity is low and there is less jostling to upset the stomach.

SYMPTOMS. High thirst, slight weight gain, salty foods taste bad, high urine output, puffy hands, and possible confusion or poor decision-making. This scenario shares some characteristics with the more dangerous overhydration and hyponatremia, including puffy hands, possible confusion, weight gain, and high urine output. The key differences are that with this scenario you will be thirsty, salty foods will taste bad, and your urine will have more color (although it is still light).

HOW TO GET BACK TO THE CENTER SQUARE. Restrict fluid intake to just moisten the mouth. Consider just swishing fluid in the mouth and spitting it out. Restrict sodium intake. If you need calories, seek low-sodium carbohydrate sources. This is not a scenario that increases risk of heat stress because there is plenty of fluid available for sweat.

COMBINATION 9: EUHYDRATED AND NORMONATREMIC

- **HYDRATION:** Normal
- **SODIUM:** Normal
- **OCCURRENCE:** Common (>30 percent of 100-mile finishers)

HOW YOU GOT HERE. You're doing everything right and adjusting fluid and sodium intake appropriately for changing conditions. Congratulations! The important thing is to stay here, which requires foresight.

SYMPTOMS. Weight is stable or slightly low, but within 3 percent of starting weight. Normal appetite and sweating, and normal urine output. Mouth is moist, and you have no nausea.

HOW TO STAY IN THE CENTER SQUARE. Think about how the conditions are going to change in the next few hours and plan accordingly. If fluid loss is likely to accelerate based on rising temperature, higher elevation, or greater intensity, plan to carry more fluid and gradually increase sodium consumption (or at least have sources of sodium with you). With so many variables in play, there are bound to be errors that take you away from the center square. While continuing to move forward, the preferable hydration and sodium errors are slight dehydration and slight hypernatremia. As you head in the direction of overhydration and/or hyponatremia, running becomes uncomfortable because of nausea, puffy hands, bloating, and a sloshing stomach. Heading in the direction of dehydration and/or hypernatremia will be increasingly uncomfortable as well, characterized by increased thirst, dry mouth, and dry skin. While these

symptoms are unpleasant, they are more easily solved (get water, wet skin to mitigate heat stress) and less likely to stop you completely.

Final Word

Your biggest takeaway from this chapter is this: sports nutrition is dynamic. There is no singular formula that will produce optimal results for all athletes in all conditions. This is an area where, as an athlete, you have to understand how to manipulate carbohydrate, fluid, and sodium based on exercise intensity, duration of activity, and environmental factors. Hopefully you now have a better understanding of how these three key ingredients interact with each other and are affected by numerous factors. Sports nutrition habits are also trainable. The first step is to determine the food and drink combinations that work best for you based on the ideas presented in this chapter. Then, over time, you can train to increase the amounts of those foods and drinks you can consume without upsetting your stomach.

KEY POINTS ON ADAPTING NUTRITION STRATEGIES FOR ULTRARUNNING

Sports nutrition strategies need to be adapted to ultra-endurance events with consideration for event duration, environmental conditions, and access to aid stations.

Athletes should develop a variety of foodstuffs that they can rely on in race situations.

Enhanced fat metabolism is a natural byproduct of endurance training.

Enhancing fat metabolism comes at a penalty in that it requires more oxygen to liberate an equivalent amount of energy as compared to carbohydrate.

Enhancing fat metabolism can be done through chronic dietary manipulation as well as acute training interventions. There are advantages and disadvantages to any of these strategies.

If athletes choose to adopt a fat-metabolism-enhancing strategy, a low-risk approach is warranted, such as doing some EnduranceRun sessions with low carbohydrate availability.

Athletes are best served to avoid extreme fat-adaptation strategies such as restricting carbohydrates on hard workouts and longer runs.

Hydration can be extremely difficult to get right and problematic to diagnose when things go wrong.

Ultramarathon nutrition is dynamic, and athletes should practice their nutrition in training multiple times under different conditions to account for variances in environmental conditions and intensity.

